INNOVECMO Project

Simulation Strategy for Liebau-Based Valveless Pump: From Structural Modeling to Reduced-Order Multiphysics Coupling

Joaquín A. Hernández

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1 Introduction

The INNOVECMO project aims at the development of a novel valveless pumping mechanism based on the Liebau principle, with the objective of reducing hemolysis and thrombosis compared to conventional rotary pumps used in ECMO systems.

The core concept relies on the generation of **asymmetric pressure waves** in a compliant tube by means of a localized actuator. These waves propagate along the tube and, due to geometric and material asymmetries, produce a net flow without valves.

The physical intuition behind this mechanism can be understood using simple analogies, such as the manual compression of a deformable container.

Figure 1: Illustrative analogy using a plastic bottle to explain wave generation and asymmetric deformation under localized compression.

Figure 2: Manual pinching of a deformable container showing localized deformation and propagation of shape changes.

These images, used during the meeting, provide an intuitive understanding of:

- Localized deformation induced by external actuation,
- Propagation of mechanical disturbances along a compliant structure,

- Sensitivity to geometry, material properties, and actuation position.

The actual experimental setup replaces the bottle with a **silicone tube connected to two reservoirs**, as discussed in the Valladolid experiments.

2 Physical Mechanism of the Liebau Pump

The system consists of:

- A compliant silicone tube,
- Two reservoirs (inlet and outlet),
- A localized actuator (pincher),
- Asymmetric tube lengths on either side of the actuator.

The key mechanism is:

1. The actuator applies a localized compression.
2. A pressure wave is generated and propagates in both directions.
3. Due to asymmetry in tube lengths and properties, the waves travel differently.
4. Phase differences and reflections produce a net directional flow.

This mechanism mimics physiological pulsatile flow and can achieve clinically relevant flow rates:

- Neonates: ~ 3 L/min
- Adults: ~ 5 L/min

3 Experimental Reference Configuration

The Valladolid team performed structural experiments on the tube without fluid to isolate deformation behavior.

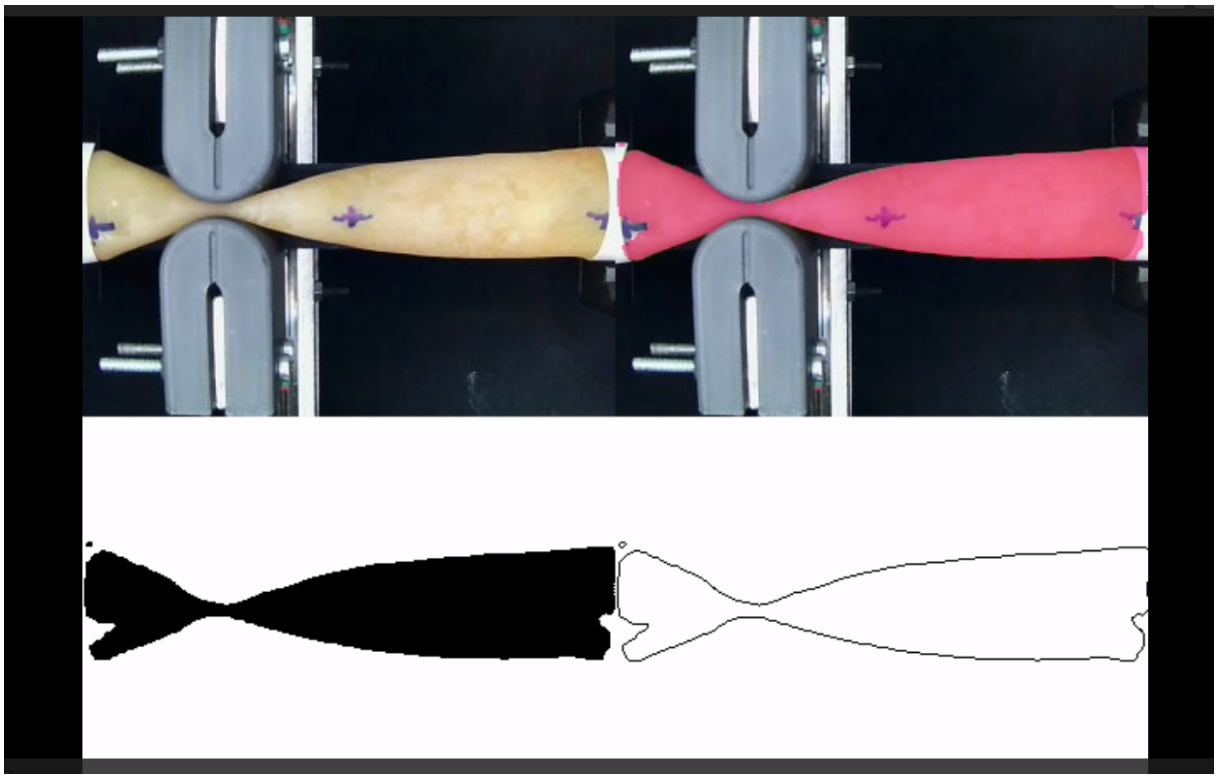

Figure 3: Experimental images of the silicone tube under localized pinching (top) and corresponding segmented contours (bottom).

Key observations:

- Large deformations and strong geometric nonlinearities,
- Progressive contact between tube walls,
- Strong dependence on actuator position,
- Availability of segmented contours for validation.

4 Modeling Challenges

The problem presents several coupled nonlinearities:

4.1 Structural Nonlinearity

- Hyperelastic material behavior (silicone),
- Large deformations,
- Instabilities (local buckling),
- Strong dependence on pre-stress.

4.2 Contact Mechanics

- Contact between tube walls,
- Moving contact regions,
- Predictable but highly nonlinear contact topology.

4.3 Fluid–Structure Interaction (FSI)

- Strong coupling between deformation and flow,
- Wave propagation phenomena,
- Inertial effects in fluid.

4.4 Geometric Evolution

The geometry evolves dynamically and directly influences flow resistance and pressure drop.

5 Pre-Stress and Initial Configuration

A critical aspect highlighted during the discussion is the role of **pre-stress**.

The tube in experiments is not stress-free:

- It is stretched between two supports,
- This generates an initial stress field,
- This defines the reference configuration.

Simulation strategy:

1. Start from an undeformed configuration,
2. Apply elongation to reach experimental length,
3. Use this as initial condition,
4. Perform dynamic simulation from this state.

Without pre-stress, the problem is ill-posed.

6 Simulation Strategy

6.1 Stage 1: Structural Model (No Fluid)

Objective: reproduce experimental deformation.

- Hyperelastic tube,
- Dynamic simulation (quasi-static insufficient),
- Imposed displacement or contact-driven actuation,
- Validation against segmented contours.

Two approaches:

- **High-fidelity contact (Abaqus)**:
  - Accurate contact resolution,
  - Used to extract contact patterns.
- **Simplified actuation (Kratos)**:
  - Imposed displacement field,
  - Gaussian or analytical approximation of pincher,
  - Faster and suitable for ROM generation.

6.2 Stage 2: Reduced Order Modeling (ROM)

Key idea: reduce computational cost.

- Train ROM using FEM simulations,
- Capture deformation as function of actuator parameters,
- Use projection-based methods (POD, etc.).

Further step:

- **Manifold ROM (M-ROM)**:
  - Capture nonlinear solution manifold,
  - Enable efficient design exploration.

6.3 Stage 3: Fluid Modeling

Two possible approaches were discussed:

6.3.1 3D Fluid Simulation

- Finite Volume (e.g., OpenFOAM),
- Structured meshes feasible,
- High computational cost,
- Limited physiological relevance.

6.3.2 1D Hemodynamic Model (Preferred)

- Already implemented in Kratos,
- Based on cross-sectional area evolution,
- Captures wave propagation,
- More relevant for cardiovascular applications.

7 Coupling Strategy: Structure + 1D Fluid

The key coupling variable is the **cross-sectional area**.

- Structural ROM $\rightarrow$ provides area evolution,
- 1D model $\rightarrow$ solves pressure and flow.

Alternatively, pressure drop can be modeled as:

$$\Delta P = K_1 Q + K_2 Q^2 + K_3 \frac{dQ}{dt} \tag{1}$$

where:

- Q is flow rate,
- K_i depend on geometry and material,
- These coefficients can be learned from simulations.

8 Multifidelity Strategy

The project is fundamentally based on a **multifidelity pipeline**:

1. High-fidelity FEM simulations (Abaqus / Kratos),
2. Reduced-order models (ROM),
3. Manifold-based surrogate models,
4. Fast design optimization.

This allows:

- Efficient exploration of actuator design,
- Optimization of flow rate,
- Reduction of hemolysis risk.

9 Software Roles

9.1 Abaqus

- High-fidelity structural simulations,
- Contact benchmarking,
- Parametric studies (batch simulations).

9.2 Kratos Multiphysics

- Structural solver with ROM capabilities,
- 1D fluid solver,
- Co-simulation framework,
- Final integrated platform.

9.3 MATLAB

- Rapid prototyping,
- Hyperelastic testing,
- Data generation.

10 Key Technical Decisions

From the meeting, the following decisions emerge:

- Start with **structural-only simulations**,
- Include **pre-stress** explicitly,
- Use **dynamic simulations** (not quasi-static),

- Prefer **simplified actuation** over full contact initially,
- Develop **ROM before full FSI**,
- Couple with **1D fluid model** rather than full 3D CFD,
- Use **Abaqus for reference** and **Kratos for pipeline**.

11 Conclusion

The meeting established a clear and pragmatic simulation roadmap:

- Begin with validated structural modeling,
- Build reduced-order representations,
- Couple with efficient 1D fluid models,
- Avoid unnecessary 3D complexity,
- Focus on design-relevant outputs.

The combination of:

- experimental data,
- high-fidelity simulation,
- reduced-order modeling,

forms a robust framework for the design of next-generation ECMO pumping systems.

A Appendix: Detailed Simulation Roadmap and Modeling Rationale

This appendix develops, in a more systematic and technically detailed manner, the simulation strategy that emerged from the discussions held with Sebastián Ares de Parga, while consistently embedding that discussion within the broader framework established during the Valladolid meetings. The purpose of this appendix is not merely to list computational tasks, but to articulate a coherent numerical narrative: what should be simulated first, why that order is justified, what physical mechanisms are to be retained at each stage, and how the final computational architecture should support the long-term objectives of the INNOVECMO project.

The central engineering problem is the design and analysis of a valveless Liebau-based pumping device, intended as an alternative to conventional rotary ECMO pumps. The Valladolid report already made clear that the ultimate mechanism of interest is not simply a local squeezing of a tube, but a *wave-driven transport process* in a compliant conduit embedded within a broader hydraulic environment. The report also emphasized that the relevant performance metrics are clinical rather than merely geometric: adult flow rates in the range of approximately 2–7 L/min, pressure rises on the order of 200–300 mmHg, and strict constraints on venous suction and blood damage indicators must be respected . At the same time, the report identified contact-dominated fluid–structure interaction and reduced-order modeling as the central computational challenges and opportunities of the project .

The informal meeting with Sebastián refined this picture in an important way. It distinguished very clearly between several levels of modeling ambition, which should not be confused. On the one hand, there is the *full physical problem*: a liquid-filled compliant tube, subjected

to localized actuation, embedded in an asymmetric circuit, producing propagating and reflected waves whose interference yields a net mean flow. On the other hand, there is the *first computational task that is realistically tractable and immediately useful*: the purely structural deformation of the empty silicone tube under the imposed action of the actuator, including large deformation, prestress, possible local buckling, and eventual wall-to-wall contact. This distinction is fundamental. The first task is not a poor approximation of the second; rather, it is the correct initial stage in a hierarchical modeling strategy. It isolates the geometric and constitutive nonlinearities of the solid, which must be understood and validated before any fluid model, whether three-dimensional or reduced, can be meaningfully coupled to it.

A.1 A.1. Physical interpretation of the problem and implications for simulation

The visual explanation given during the meeting, including the bottle analogy and the segmented experimental images, is particularly revealing. The bottle images serve as a pedagogical representation of a broader idea: a compliant container subjected to localized external compression does not merely deform locally, but redistributes internal pressure and geometry over a finite region. In the actual device, the same concept appears in a more controlled form. A silicone tube, connected to two reservoirs and arranged asymmetrically with respect to the actuation point, is compressed by a pinching mechanism. The resulting disturbance travels in both directions; because the lengths and impedances of the two sides are not equal, the reflected waves are not synchronized. This lack of symmetry is precisely what allows net pumping.

From the viewpoint of simulation, several consequences follow.

First, the problem is *inherently transient*. Even in the absence of fluid, the structural response of such a thin and highly compliant tube is not expected to be well captured by a purely quasi-static viewpoint once localized flattening and geometric instabilities appear. The discussion stressed that local micro-buckling may occur if the wall is sufficiently thin, and that attempting to enforce a purely static solution path may be numerically fragile or even misleading. This is entirely consistent with the general theory of nonlinear thin structures, where instability-sensitive configurations often require dynamic or pseudo-dynamic continuation even when the physical loading is slow..

Second, the problem is *prestress-dependent*. The experimental tube does not begin in a natural stress-free state. It is mounted between supports and held stretched, which means that the configuration observed before actuation already contains axial tension. The meeting repeatedly emphasized that this is not a secondary detail: the apparent straightness of the tube is itself a consequence of prestretch. Therefore, any simulation that starts from an unstretched tube and applies a pinching action directly is physically inconsistent. The correct simulation workflow must begin by constructing an equilibrium reference state under axial elongation, and only then apply the actuator motion. This mirrors standard practice in arterial and soft-tissue mechanics, where physiological configurations are often prestressed and the stress-free geometry is not the correct computational reference for dynamic response [**?**, **?**].

Third, the problem is *contact-sensitive but geometrically predictable*. The contact between the external actuator and the tube wall is not an arbitrary impact problem; nor is the eventual wall-to-wall contact inside the tube completely unconstrained in its location. The discussion pointed out that the contact pattern is, to a large extent, foreseeable from the geometry of the pincher. This observation is important because it opens the door to two distinct numerical strategies: one may either resolve contact explicitly with a high-fidelity formulation, or one may exploit the predictability of the contact zone to construct simplified actuation models that reproduce the same geometric effect at lower cost.

Finally, the problem is *multiscale in its modeling needs*. A fully resolved three-dimensional fluid–structure simulation of the entire device and external circuit would undoubtedly be impressive visually, but this was correctly identified during the meeting as potentially excessive

from the physiological and design standpoint. What is clinically relevant is the pressure-flow-wave behavior of the whole system, not necessarily a pointwise three-dimensional description of every fluid streamline. This is why the long-term direction discussed in the meeting—a coupling between a reduced structural model and a one-dimensional wave-propagation model—is so compelling. It retains the geometric richness where it matters, namely in the deforming actuation zone, while representing the surrounding circuit with a model that is much better aligned with cardiovascular wave transmission .

A.2 A.2. Why the simulation program must be staged

The Valladolid report already proposed a phased strategy, from structural-only analysis to local FSI and then to system-level wave effects . The discussion with Sebastián confirms that this ordering is not merely convenient but methodologically necessary.

There are at least four reasons for this.

The first is *identifiability.* If one begins immediately with a coupled FSI simulation, then any discrepancy with the experiment becomes ambiguous: it could come from the structural constitutive law, from the prestress level, from the contact treatment, from fluid discretization, from coupling errors, or from boundary-condition mismatch. By contrast, if one first matches the experimentally observed dry deformation profiles of the empty tube, then the structural model becomes a calibrated and validated subsystem. This reduces uncertainty dramatically before fluid is introduced.

The second is *computational economy.* High-fidelity contact-dominated three-dimensional simulations are expensive. If fluid is also included from the beginning, the offline training burden for ROM construction becomes unnecessarily large. The project was explicitly sold as a multifidelity strategy: a relatively small number of high-fidelity simulations should feed reduced models, and those reduced models should in turn support more extensive design-space exploration. This logic strongly favors an initial structural stage in which the complexity is already substantial, but still sufficiently limited to permit systematic sampling of actuator positions, amplitudes, and perhaps geometric variants.

The third is *software realism.* The meeting made clear that Abaqus can be used effectively for high-resolution contact studies and automated batches, whereas Kratos is the natural platform for projection-based reduced modeling, one-dimensional hemodynamic formulations, and co-simulation. It would therefore be unwise to collapse all objectives into a single first implementation. A better strategy is to use each tool where it is strongest in the early phases, and only later consolidate the reduced workflow inside Kratos.

The fourth is *scientific communicability.* As discussed informally, the project is coordinated and therefore cannot rely solely on broad future promises. Tangible intermediate outputs are needed. A first validated structural simulation of the tube under imposed actuation, especially if it can be compared against segmented experimental contours, already constitutes a credible and communicable result. It provides figures, validation plots, and a concrete technological demonstration that can support presentations, internal reviews, and dissemination, even before full FSI is available.

A.3 A.3. Stage I: high-fidelity structural model of the empty tube

The first simulation stage should therefore consist of a three-dimensional structural analysis of the silicone tube in the absence of internal fluid. The objective is not to infer pumping performance, but to reproduce the experimentally observed deformation patterns under the action of the pincher.

Geometry and boundary conditions. The tube should be modeled with its actual dimensions, wall thickness, and end constraints. The experimental setup suggests that the tube is

mounted between two supports and held under tension. Thus, the computational boundary conditions should be constructed in two substeps. In a first substep, an initially shorter and/or slightly larger-diameter tube is elongated until the observed mounted length is reached. This generates a prestressed configuration. In a second substep, the actuator motion is applied to this prestressed state. Conceptually, this is preferable to prescribing an ad hoc initial stress field, because it produces a mechanically consistent reference state.

Constitutive law. At this stage, a hyperelastic constitutive model is sufficient and appropriate. The report and the meeting both point toward silicone-like large-deformation behavior without plasticity. This is fortunate, because the project's manifold-based nonlinear reduction technology is much more mature in the hyperelastic setting than in the plastic one. Depending on the material data available from Valladolid, a Neo-Hookean, Mooney–Rivlin, or more general incompressible hyperelastic law may be used as a first approximation. The essential point is not initially to identify the most refined constitutive model, but to have a stable large-strain formulation consistent with the calibration information already available from the Valladolid group.

Actuation model. Here the discussion opened two paths. One is to model the actuator explicitly as rigid bodies pressing the tube, with contact. The other is to replace the actuator by an imposed displacement field or moving geometric indentation that reproduces the same effect. Both are useful, but for different purposes.

The explicit contact model is more physically transparent and should serve as the reference case, especially in Abaqus, where high-resolution contact studies can be automated. Such simulations can reveal the actual contact footprint and the transition from first touch to near-closure. However, for reduced-order model generation, it may be advantageous to replace explicit contact by a smoother surrogate actuation law, for instance one derived from the actuator geometry or from a fitted indentation profile. The meeting suggested that for the cylindrical pincher currently used, the contact is sufficiently predictable that such a surrogate may be both feasible and effective. This is a key idea: one may use a small number of high-fidelity contact simulations to infer the geometric pattern of indentation, and then build a larger structural training set using that surrogate loading.

Dynamics versus quasi-statics. The structural stage should not be posed naively as a purely quasi-static problem. Because the tube is thin and highly compliant, slow compression may still trigger local instabilities that are numerically difficult to follow in a static framework. A dynamic implicit or weakly damped transient formulation is therefore advisable, even if the physical actuation is slow. This is not intended to reproduce true inertial structural oscillations in detail, but to regularize the nonlinear solution path and permit stable traversal of snap-through-like configurations or local wall flattening events. The meeting explicitly recognized this issue, and it is consistent with standard nonlinear computational practice.

Validation target. The segmented images are extremely valuable here. Rather than validating only scalar quantities, the first simulations should be compared against extracted tube profiles. A coherent validation protocol could involve, for each actuator position and imposed displacement:

1. comparison of the projected outer contour,
2. comparison of local tube width along the centerline,
3. comparison of throat width and closure onset,
4. comparison of contact or near-contact region length.

This image-based validation is especially powerful because it tests geometry directly, which is precisely the quantity that will later be passed to the one-dimensional fluid model.

A.4 A.4. Role of Abaqus in the early phase

Although the strategic computational platform of the project is Kratos, the discussion correctly recognized that Abaqus can play a practical role in the early phase.

Abaqus is particularly well suited for:

- rapid setup of large-deformation hyperelastic structural simulations,
- robust handling of contact,
- high-resolution reference runs,
- systematic batch studies through scripting.

This does not mean that Abaqus should become the core of the project. Rather, it should be treated as a *reference engine* for generating trustworthy snapshots and for clarifying uncertain aspects of the contact mechanics. For instance, one may perform a family of simulations varying the actuator position, actuator radius, indentation depth, or tube length asymmetry, and then use the resulting deformation fields either directly as training data or indirectly to infer simplified actuation patterns for Kratos.

This is entirely compatible with the multifidelity philosophy stated in the project. The role of the expensive solver is not to explore the entire design space exhaustively, but to anchor the reduced hierarchy in a set of high-quality examples.

A.5 A.5. Role of Kratos in the early and intermediate phases

Kratos should be the main environment for building the project's long-term modeling chain. The discussion identified several reasons for this.

First, Kratos already supports reduced-order techniques and co-simulation, which are essential to the project objectives. Second, the geometry of the current structural problem is sufficiently simple that implementing the tube model in Kratos should not be prohibitive. Third, once the structural model is in Kratos, it can evolve naturally into the ROM, manifold-ROM, and one-dimensional coupling stages without having to abandon the software environment.

The correct conceptual use of Kratos is therefore not merely to reproduce a single structural simulation, but to create a *computational scaffold* on which the later phases will be built. In practical terms, this means that the first Kratos model should already be designed with future reduction in mind: clean parameterization of actuator motion, modular handling of prestress, storage of snapshots, and extraction of geometry-dependent quantities such as local cross-sectional measures.

A.6 A.6. Stage II: projection-based structural ROM

Once the full-order structural model is reliable, the next step is to construct a reduced-order model of the deforming tube. At this stage, the reduction need not yet be manifold-based. A classical projection ROM, built from snapshots of displacement and possibly stress or internal-force data, is a natural intermediate level.

The rationale is straightforward. The parameter space is not enormous initially. The actuator position, indentation amplitude, tube prestretch, and perhaps one or two geometric parameters may suffice as first coordinates. A carefully designed training set spanning these parameters should yield a structural response manifold of moderate intrinsic dimension. Therefore, a projection-based ROM may already provide substantial acceleration for interpolation within this initial design envelope.

It is important to understand what this ROM is for. It is *not* the final surrogate for optimization, at least not in the ambitious version of the project. Rather, it is the first compressed representation that still remains close to the mechanics and can be coupled to other models. It may be used in at least three ways:

1. to accelerate additional structural simulations,
2. to support a first proof-of-concept structural co-simulation with fluid,
3. to generate abundant data for the later manifold reduction and surrogate stages.

During the meeting, Sebastián correctly distinguished between a standard ROM and a manifold reduced model. That distinction is important. A projection ROM remains linear in its reduced ansatz, even if the original mechanics are nonlinear. A manifold model aims to encode the nonlinear geometry of the solution set more directly. But from a workflow standpoint, the standard ROM is an indispensable stepping stone.

A.7 A.7. Stage III: manifold reduced model and multifidelity hierarchy

The project proposal explicitly committed to a multifidelity strategy in which a small number of expensive finite element simulations would be used to construct a reduced model, and that reduced model would then support an even cheaper manifold-level representation for design tasks. The informal meeting reaffirmed that this is still the correct long-term vision.

The logic of the hierarchy may be stated as follows.

At the top sits the *high-fidelity structural model*, whose role is accuracy and physical trustworthiness. Below it sits the *projection ROM*, whose role is still mechanics-based simulation but at lower cost. Below that sits the *manifold-level surrogate*, whose role is design exploration, parametric screening, and rapid evaluation.

This hierarchy is not redundant. Each level serves a different purpose:

- The full model establishes truth.
- The ROM transfers truth into a compressed simulation environment.
- The manifold surrogate transfers compressed simulation into design speed.

For the present problem, this is especially attractive because the tube deformation is controlled by relatively few physically meaningful inputs, while the response is strongly nonlinear due to contact, flattening, and geometric stiffness. Such problems are natural candidates for nonlinear manifold learning and hyper-reduction strategies, provided that the training data are informative and that the reduction respects the structure of the mechanics .

A.8 A.8. Why a full 3D fluid simulation is not the right long-term centerpiece

One of the most insightful points made during the meeting is that a full three-dimensional fluid model of the whole device, although feasible in principle, may be the wrong centerpiece for the physiological aims of the project.

This is not because 3D CFD is unimportant. On the contrary, a local 3D fluid calculation may later prove useful for understanding details of shear, recirculation, or hemolysis. But the primary clinical effect of the Liebau-type device is not a local vortex pattern inside a short tube segment; it is the generation, reflection, and interference of waves over an extended hydraulic system. These are naturally one-dimensional phenomena to leading order. Indeed, one-dimensional hemodynamic models are standard precisely because they retain the dominant pulse-wave physics over long compliant vessels at a fraction of the cost of 3D formulations .

Thus, the correct long-term question is not: *Can we solve the entire fluid problem in 3D?* The correct question is: *What is the minimum fluid model that preserves the wave and impedance mechanisms relevant to the pump, while still allowing the local geometric effects of actuation to enter accurately?* The answer discussed in the meeting is highly plausible: a one-dimensional wave-propagation model, coupled to a local reduced structural model of the actuated zone.

A.9 A.9. Stage IV: coupling the structural ROM with a 1D fluid model

The coupling idea that emerged during the discussion is elegant and should be made central in the project narrative.

A one-dimensional fluid model evolves pressure and flow along the conduit using cross-sectional area as a key state variable. The structural reduced model, in turn, determines how the tube geometry changes under actuation. Therefore, the reduced structural model does not need to provide a full three-dimensional fluid domain to the hemodynamic solver; it needs to provide the quantities that the one-dimensional model actually uses. Chief among them is the effective local cross-sectional area, or more generally a local constitutive relation between geometry and hydraulic resistance.

Two coupling philosophies were discussed.

The first is *direct geometric coupling.* In this approach, at each time instant, the structural reduced model provides the local area evolution induced by the actuator. The one-dimensional solver then propagates pressure and flow accordingly. This is a geometrically transparent approach and likely the most natural if the 1D solver is already formulated in terms of area and pressure.

The second is *reduced hydraulic closure.* In this approach, one does not pass area directly, but rather identifies an effective pressure-drop law of the form

$$\Delta P = K_1 Q + K_2 Q^2 + K_3 \dot{Q}, \tag{2}$$

where the coefficients K_1, K_2, and K_3 depend on the instantaneous deformed geometry generated by the actuator. In other words, the structural model parametrizes the local hydraulic law, and the one-dimensional solver uses that law as a localized device model. Sebastián remarked that a related idea had already been used in the context of coronary stenoses, where a high-dimensional flow problem was compressed into a few coefficients embedded in a 1D model. The novelty here is that these coefficients become time-dependent because the geometry itself is actuated dynamically.

Both philosophies are valuable, and the project could explore both. Direct area coupling is more geometrically faithful; reduced hydraulic closure may be more compact and computationally convenient. The choice should ultimately depend on robustness, ease of calibration, and how easily each approach integrates into the existing 1D Kratos framework.

A.10 A.10. Stage V: local FSI and the role of fluid snapshots

The meeting also kept open the possibility of a more explicit local FSI simulation, especially as a proof of concept. This should be interpreted carefully.

A local FSI calculation—whether with a full structural model or, more ambitiously, with a reduced structural model—could be extremely valuable as an intermediate scientific result. It would demonstrate that the solid can indeed be reduced while retaining two-way coupling with the fluid. Such a result would be entirely aligned with the project's claim that reduced-order structural technology can be used inside FSI settings.

However, it would be a mistake to let this local FSI stage consume the whole effort. Its role should be demonstrative and methodological, not to replace the final physiologically meaningful

system-level model. In other words, local FSI is scientifically attractive and publishable; 1D-coupled wave modeling is physiologically central. Both belong in the project, but they should not be confused.

A.11 A.11. Suggested practical sequence of actions

Based on both the Valladolid report and the informal discussion, the following practical sequence appears to be the most coherent.

Step 1: collect and consolidate the reference data. Before any substantial implementation, the exact tube geometry, wall thickness, actuator geometry, material parameters, and available ANSYS or experimental files from Valladolid should be gathered and organized. The meeting explicitly noted that the Valladolid group already has models and parameter sets, possibly including ANSYS data and calibrated silicone properties. This material is essential to avoid reinventing a reference case from scratch.

Step 2: build a prestressed structural tube model in Kratos. A first Kratos model should be created with the sole objective of reproducing the mounted tube in its prestretched state. This model should be validated internally by checking that the final length and diameter are consistent with the experimental configuration. No actuator is needed yet.

Step 3: introduce a first actuation model without explicit fluid. A time-dependent imposed displacement or simplified contact surrogate should then be applied. The goal is to reproduce one or two experimentally observed deformed shapes. This stage should remain as simple as possible while still matching the segmentation data.

Step 4: perform a small set of high-resolution Abaqus runs. In parallel, one should produce a benchmark family of Abaqus simulations with explicit contact and possibly variations in actuator location and indentation. These runs should serve as a high-fidelity reference for contact footprint and deformation accuracy.

Step 5: compare structural simulations against segmentation. The structural validation should not be anecdotal. A systematic geometric comparison procedure should be defined, so that the simulations are anchored quantitatively to the observed profiles.

Step 6: generate structural snapshots and construct a projection ROM. Once the structural full-order model is stable, snapshot generation should begin. The parameter set may initially be small, but it should already be chosen with future design applications in mind.

Step 7: identify the interface quantities required by the 1D fluid model. In parallel with ROM construction, the exact information that the 1D Kratos model needs from the structure must be formalized: area profile, local impedance, pressure-drop coefficients, or some combination thereof.

Step 8: test a first co-simulation architecture. A simple structure–1D coupling should be attempted, ideally first in a one-way or weakly coupled manner, in order to verify consistency of data transfer and time integration.

Step 9: build the manifold-level multifidelity layer. Only after the structural ROM and the coupling logic are stable should the manifold-level surrogate be constructed for design exploration. At this point, the project's original multifidelity promise becomes operational.

Step 10: revisit local FSI as a methodological validation case. A local FSI benchmark using the reduced structural model may then be developed as a high-value methodological output, but without displacing the central role of the 1D-coupled device model.

A.12 A.12. What should be considered a successful first milestone

A successful first milestone should not be defined too ambitiously. It is not necessary, in the immediate term, to demonstrate full pumping of 5 L/min in a patient-relevant circuit. A much more realistic and scientifically sound first milestone would be:

> A validated three-dimensional hyperelastic simulation of the prestressed empty tube under localized actuation, reproducing experimentally observed deformed profiles and serving as the basis for structural reduced-order modeling.

This milestone would already be of high value. It would demonstrate that the project has moved beyond conceptual discussion and into physically grounded computational realization. It would also provide the indispensable substrate for everything that follows.

A.13 A.13. Final synthesis

The main insight emerging from the combined reading of the Valladolid report and the informal discussion is that the project should resist the temptation of premature maximalism. The full problem is unquestionably multiphysics, multiscale, and contact-dominated. But the correct way to solve such a problem is not to attack all layers at once. It is to organize them in a hierarchy.

The hierarchy that emerges is both clear and technically defensible:

1. establish the structural truth of the empty prestressed tube under actuation,
2. compress that truth into a reduced structural representation,
3. couple that reduced structural representation to a one-dimensional wave model,
4. use the resulting coupled reduced model as the engine of multifidelity design.

This is not a retreat from the original ambition of the project. On the contrary, it is the most credible route toward fulfilling it. It preserves the essential physics of the device, exploits the strengths of Kratos in reduced and coupled modeling, uses Abaqus pragmatically where it is strongest, and aligns the computational development with both experimental validation and physiological relevance.